

	Work Instruction	Doc. Revision	00
	AXI V810 STD Valve Assembly Removal & Replacement Procedure	Effective Date	25 May 2017

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☐			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	25 May 2017	First Release	Chin Chee Yan

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Removal Procedure

1. **Shutdown and completely power off machine.** Turn off air pressure regulator
2. **Power off the SSCA main power # Important**
3. Open front access door and manual move XY stage towards front door direction

Remark:

- Equipment SN MY10133095 & below equip with Inch Size Inlet Port at Valve Assembly (Figure 1)
- Equipment SN MY10133096 & above equip with Metric Size Inlet Port at Valve Assembly (Figure 1)

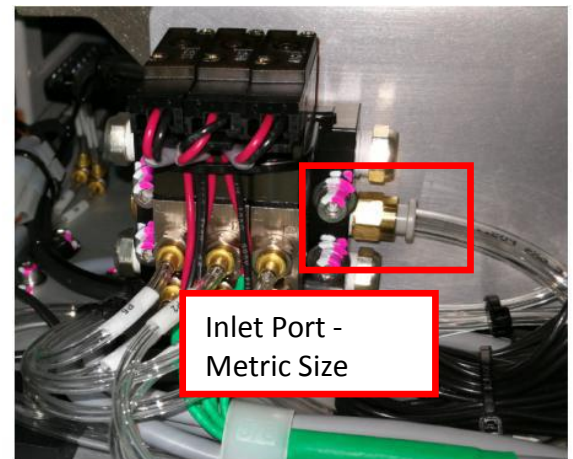
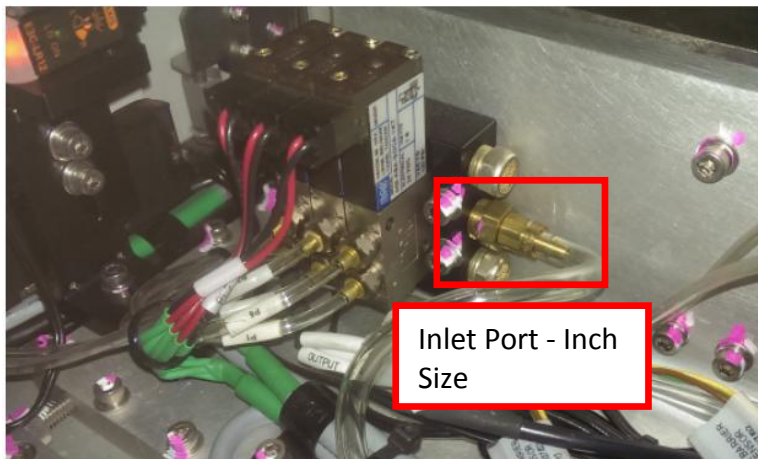


Figure 1 Inch and Metric Inlet Port

4. Remove Main Tubing by refer to below step
 - a. Inch Inlet Port Type
 - i. Remove the Clamp Hose in advance. Remove the Main Tubing

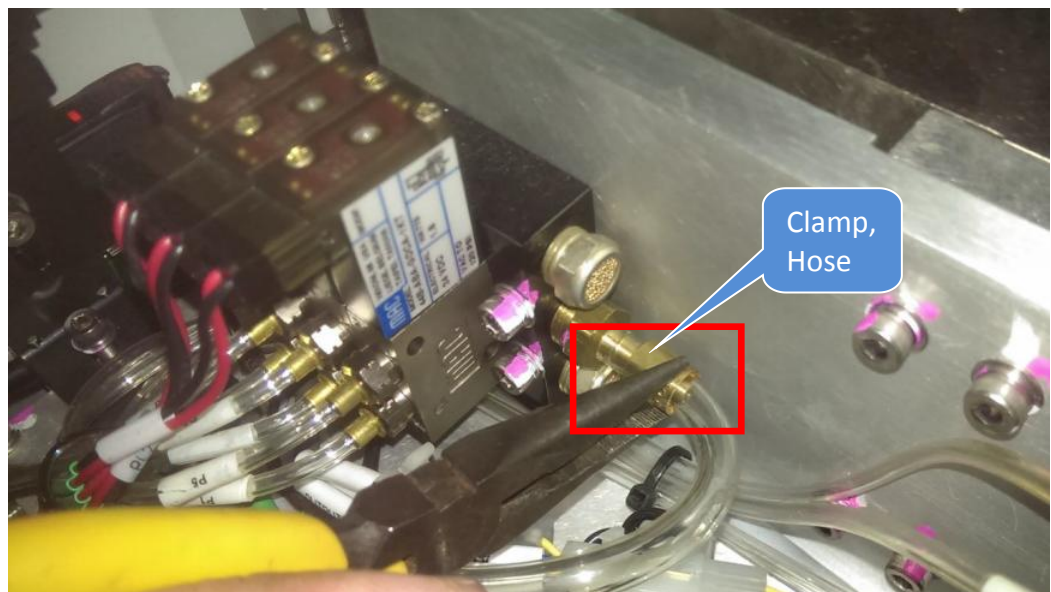


Figure 2 Clamp Hose Removal

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b. Metric Inlet Port Type

i. Press inlet port and remove the main tubing

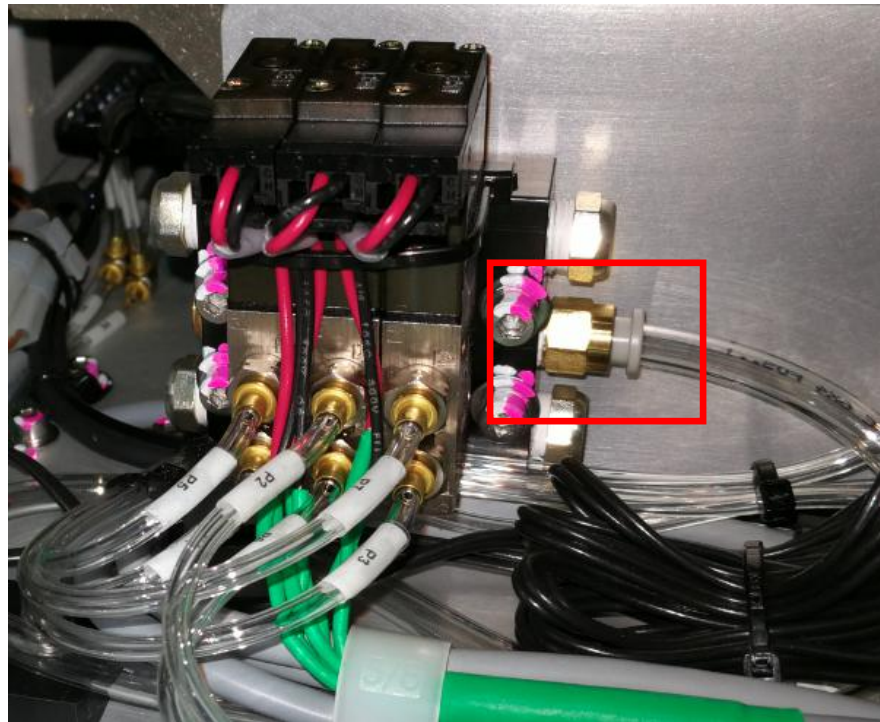


Figure 3 Main Tubing Removal

5. Disconnect 3 x Solenoid Valve Cables as below Figure 4

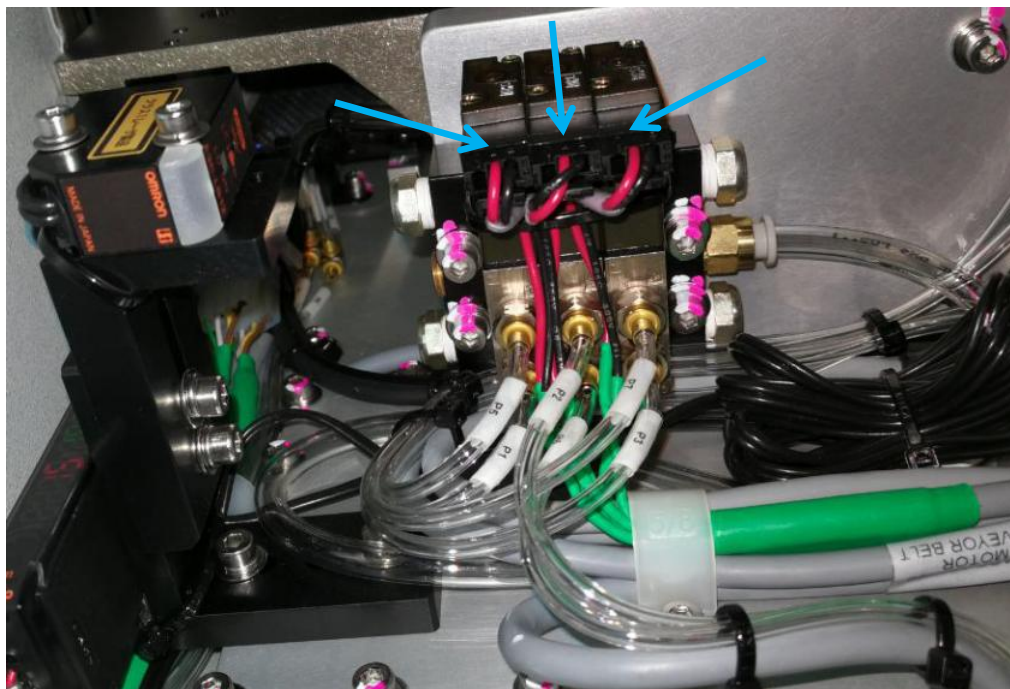


Figure 4 Solenoid Valve Cables Removal

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6. Remove 6 x Clamp Hose as below Figure 5. Remove tubing.

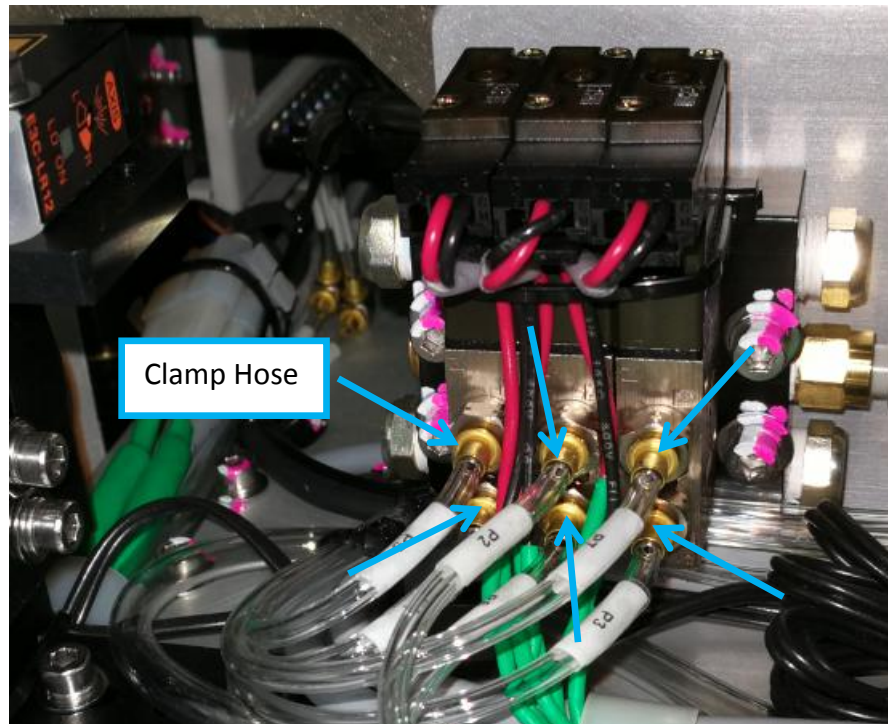


Figure 5 Clamp Hose & Removal

7. Remove 4 x Screws as below Figure 6. Remove entire Valve Assembly from Fix Rail



Figure 6 Valve Assembly Removal

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Replacement Procedure

8. Prepare new Valve, Assembly (P/N, 2830-0002) as shown.



Figure 7 Valve Assembly

9. Perform Fitting Transplant by refer to below :
 - a. Equipment SN MY10133095 & below equip with Inch Size Inlet Port at Valve Assembly
 - i. Direct refer to Step 11
 - b. Equipment SN MY10133096 & above equip with Metric Size Inlet Port at Valve Assembly (Figure 1)
 - i. Remove the Metric Inlet Port from Valve Assembly (Previously dismantle)



Figure 8 Valve Assembly (Metric Inlet Port Removal)

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- ii. Remove the Inch Inlet Port from Valve Assembly (New Spare Part)



Figure 9 Valve Assembly (Inch Inlet Port Removal)

- iii. Assemble the Metric Inlet Port at New Spare Valve Assembly as below Figure 10

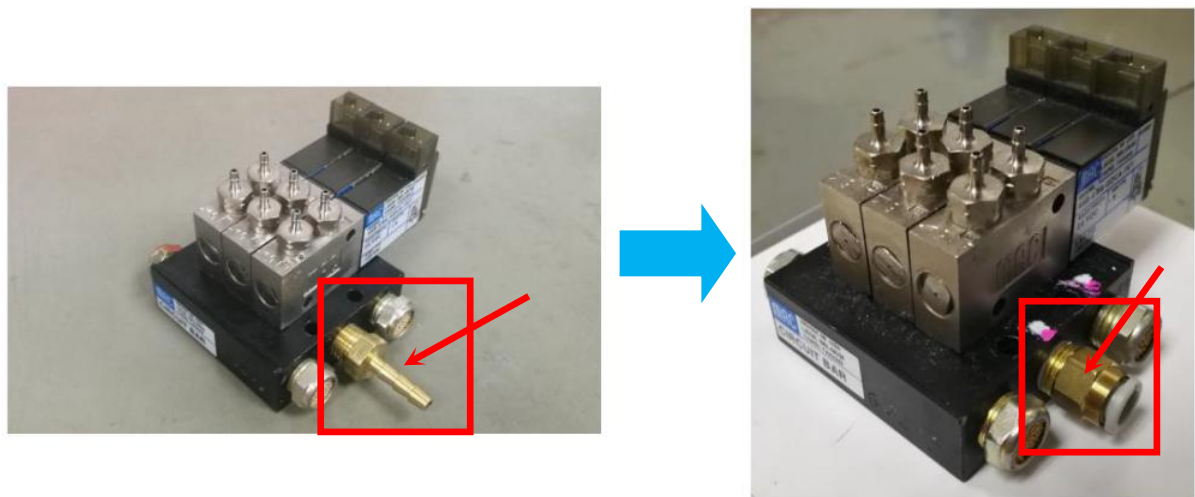


Figure 10

10. Assemble Valve Assembly at Fix Rail. Tighten 4 x Screws.



Figure 11 Valve Assembly

11. Connect 3 x Solenoid Valve Cables respectively by refer to Cable Labeling. May refer to Table 1 for reference

Valve Assembly			
Solenoid Valve Cable	PL1	PL2	PL3

Table 1

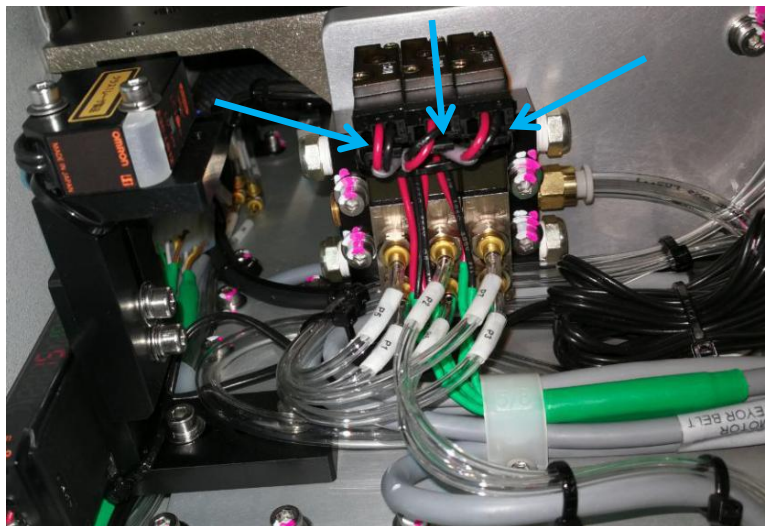


Figure 12 Solenoid Valve Cable Configuration

12. Connect 6 x Tubing Respectively by refer to Tubing Labeling. May refer to Table 2 for reference

Valve Assembly			
Solenoid Valve Cable	PL1	PL2	PL3
Tubing	P5	P2	P7
Tubing	P1	P6	P3

Table 2

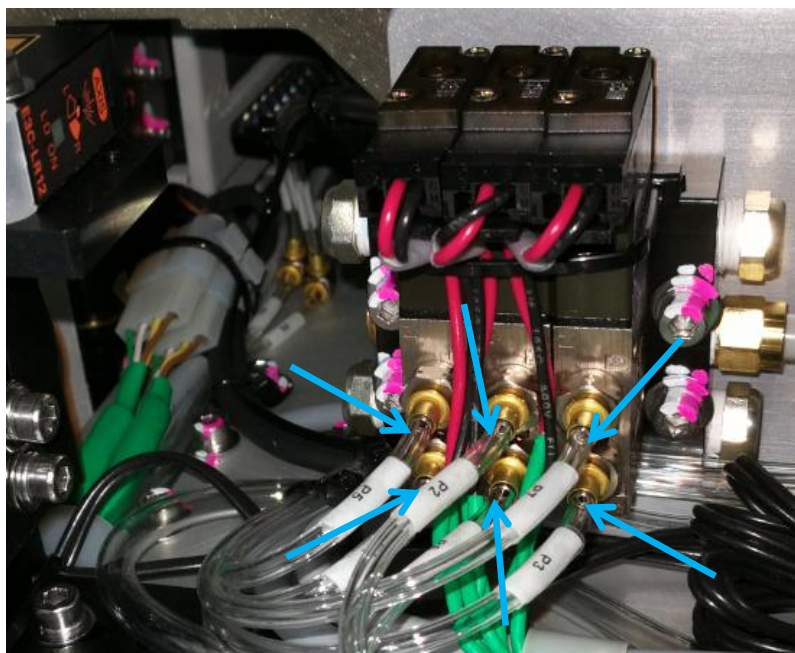


Figure 13 Tubing Assembly

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13. Connect 6 x Clamp Hose (PN : 2820-0002) as below Figure 14

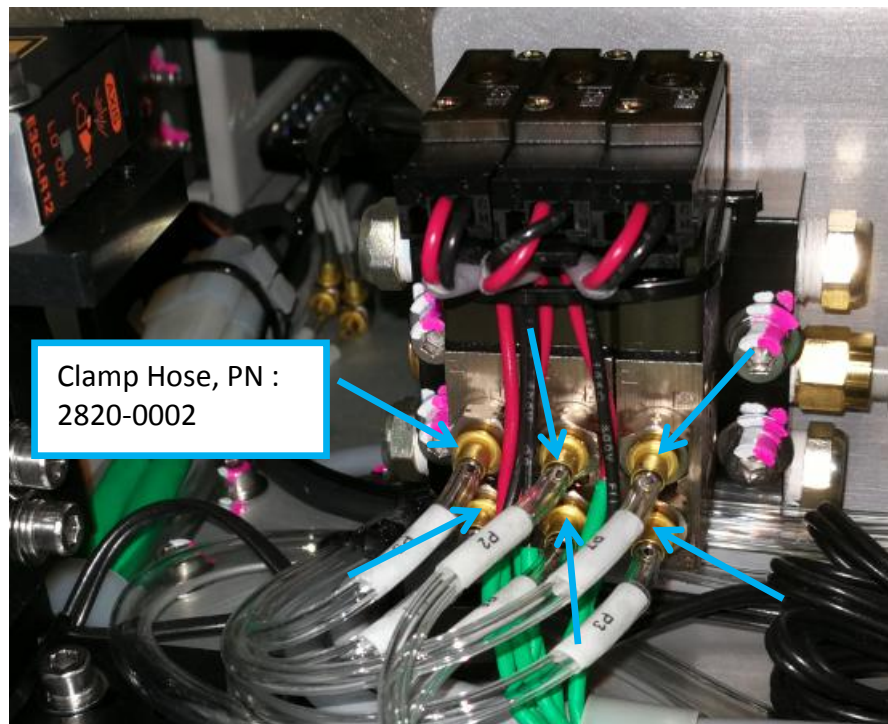


Figure 14 Clamp Hose Assembly

14. Connect Main Tubing by refer to below step

a. Inch Inlet Port Type

- i. Plug tubing into Inch Inlet Port accordingly. Next, Gently slot in Clamp Hose to secure the tubing

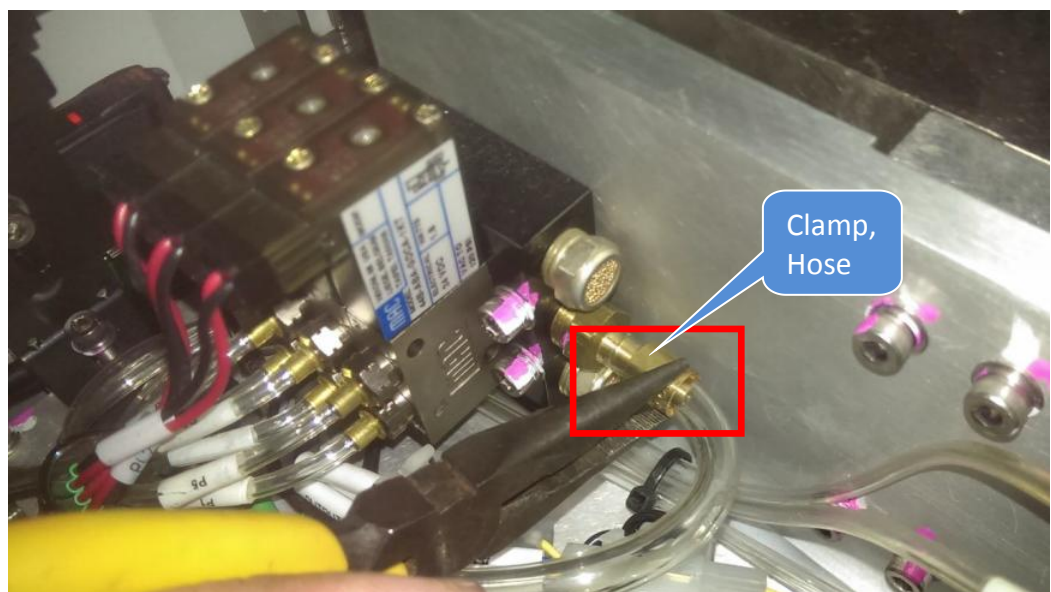


Figure 15 Clamp Hose Removal

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b. Metric Inlet Port Type

i. Plug the main tubing into Metric Inlet Port

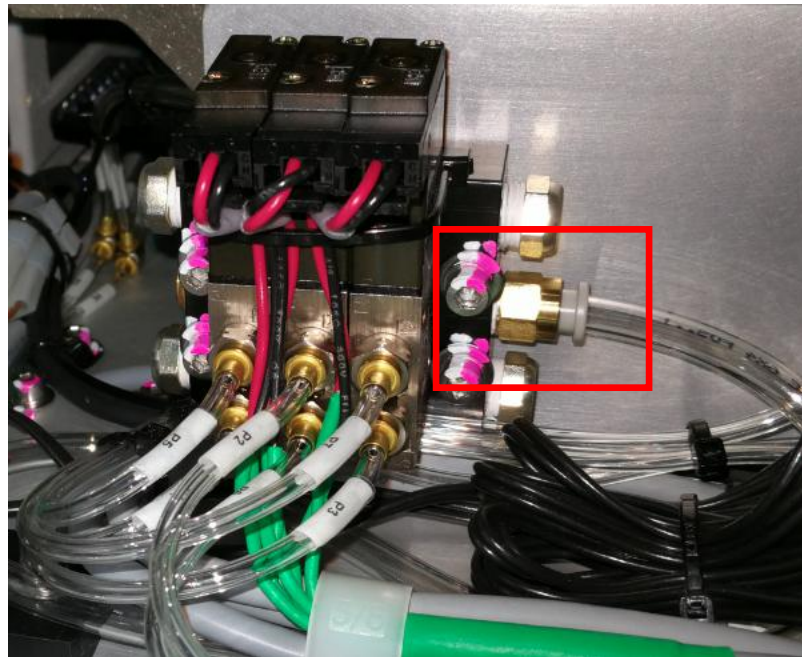


Figure 16 Main Tubing Removal

15. Gently move X axis stage and visual check where Tubing & Solenoid cable do not contain friction with E-Chain Guard Plate.

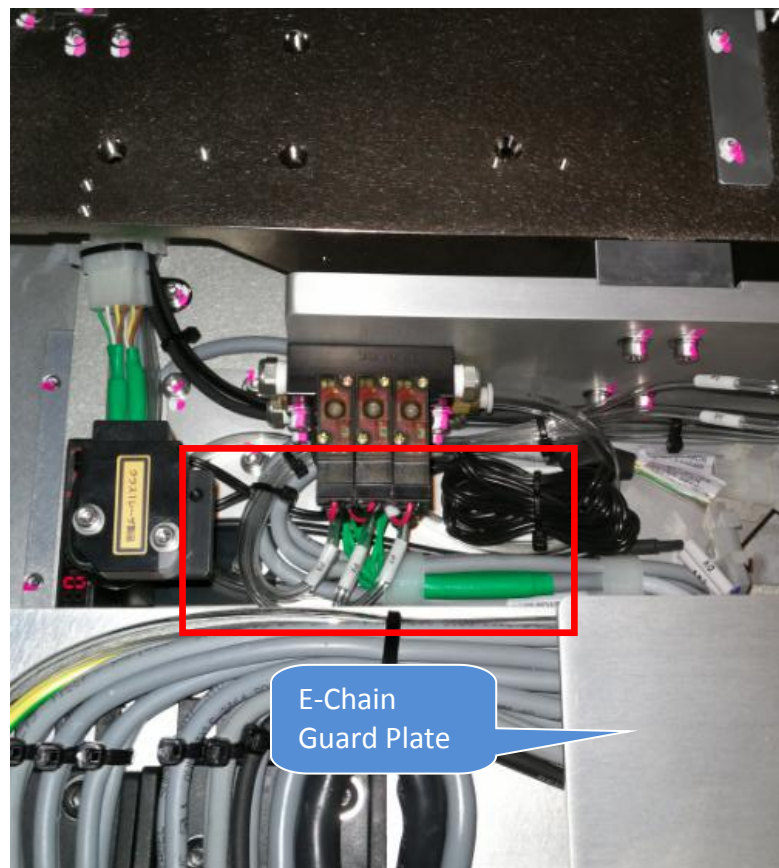


Figure 17

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16. Remove all tools from machine, turn on main air regulator and start-up machine as usual.

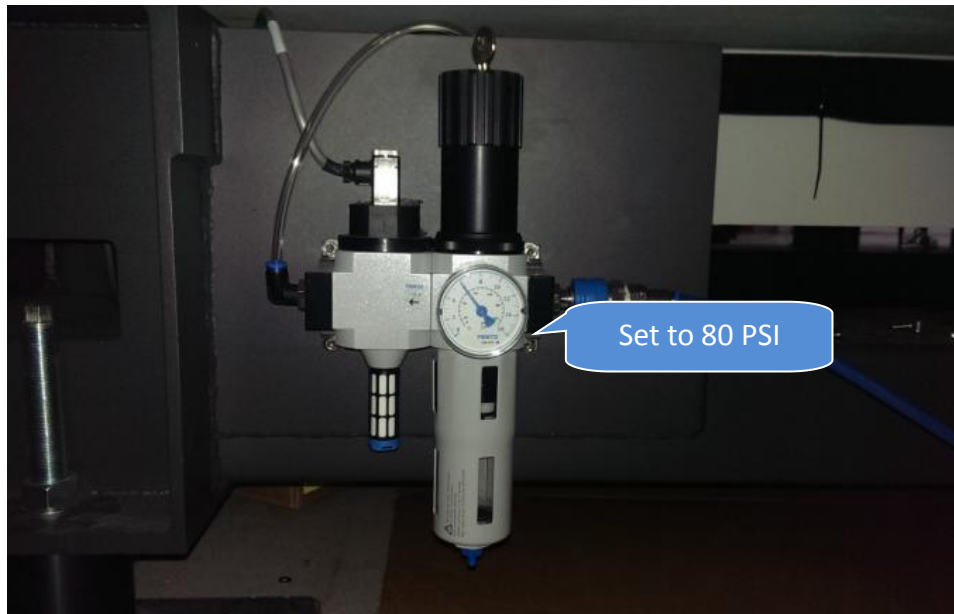


Figure 18 Air Pressure Regulator adjustment

Validation Procedure

17. Log in V810 GUI, go to panel handling and click initialize. Turn off X-ray and open machine front access door. By refer to Step E,F, G, visual check on mechanism movement, ensure actual movement is sync with the command toggle.

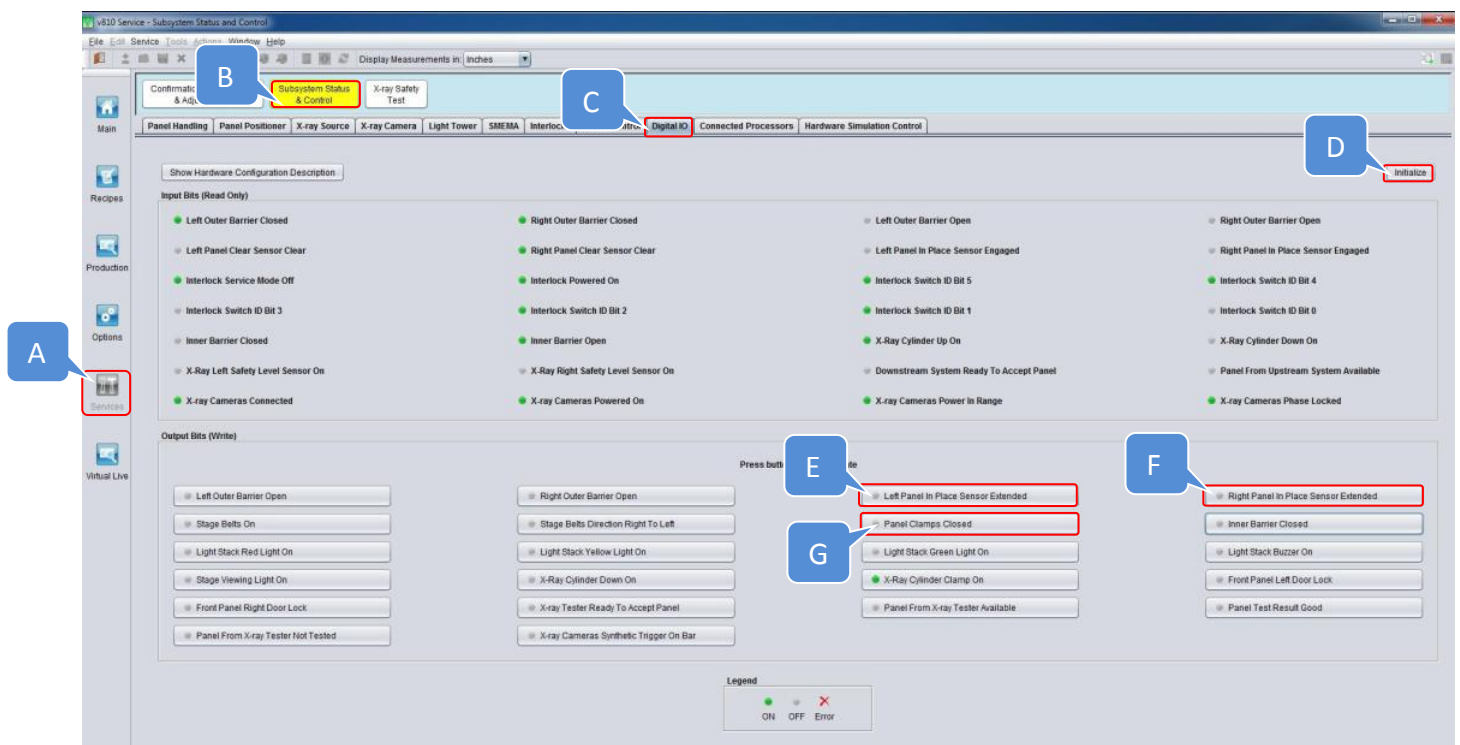


Figure 19 Panel Handling Mechanism Validation (Digital IO)

18. Close front access door and start-up as usual.